

secondary transformer (U1), a bridge rectifier, a 5V voltage regulator 7805CV, IC1 and other essential components. A ready-made 5V adaptor can alternatively be used instead of the transformer, bridge rectifier, and voltage regulator.

The device operates on a 5V DC supply, with capacitors C1 and C2 filtering any ripples.

The Arduino Nano serves as the controller. The PIR motion sensor detects human movement and transmits data to the Arduino. The PIR sensor has three pins: VCC, GND, and OUT. A 5V single-pole, double-throw (SPDT) relay used to control the load (230V AC, 60W bulb) acts as a switch. The 5V power supply is connected to the 5V (Vin) and GND pins of the Arduino Nano to enable the circuit. The VCC and GND are shared by the Arduino Nano, the PIR sensor module, and the relay, as illustrated in Fig. 2.

The OUT pin of the PIR sensor connects to digital pin 10 (D10) on the Arduino Nano. Pin 9 (D9) connects to the base of the relay driver transistor BC547 through resistor R2 and LED1.

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

sketch_oct17a.ino

```
1 int rly1 = 9;
2 int sensor = 10; // the pin that the sensor is attached to
3 int state = LOW; // by default, no motion detected
4 int val = 0; // variable to store the sensor status (value) void
5 pinMode(sensor, INPUT);
6 pinMode(rly1, OUTPUT);
7 digitalWrite(rly1, LOW);
8 // initialize RELAY as an output
9 // initialize sensor as an input
10 Serial.begin(9600);
11 Serial.println("Latif Khan");
12 Serial.println("Yogesh Jatav");
13 }
14 void loop(){
15 val = digitalRead(sensor); // read sensor value
16 if (val == HIGH) { // check if the sensor is HIGH
17 digitalWrite(rly1, HIGH); // turn RELAY ON
18 delay(60000); // delay
19 if (state == LOW) {
20 Serial.println("Motion detected!");
21 state = HIGH; // update variable state to HIGH
22 }
23 delay(60000);
24 }
25 else {
26 digitalWrite(rly1, LOW); // turn RELAY OFF
27 if (state == HIGH){
28 Serial.println("Motion stopped!");
29 state = LOW; // update variable state to LOW
30 }
31 }
32 }
```

Fig. 3: Code snippet of the source code

The Arduino Nano processes inputs from the PIR motion sensor. When motion is detected, the sensor signals the Arduino Nano, which triggers the relay and activates the bulb. The relay acts as a switch

BILL OF MATERIALS

Components	Quantity
Arduino Nano (MOD1)	1
PIR sensor (S1)	1
230V AC to 9V, 250mA secondary transformer (U1)	1
5V voltage regulator 7805CV (IC1)	1
Ceramic disc capacitor 100nF (C1)	1
Electrolyte capacitor 1000µF, 40V (C2)	1
5mm LED (red, green colour)	2
Resistors 330 ohm (R1, R2)	2
5V SPST relay (RLY 1)	1
5V DC adaptor (optional)	1
230V AC 60W bulb (as a load for testing)	1

to enable or disable the connected 230V AC, 60W bulb according to the motion sensor's input.

The Arduino software platform is used to develop the source code. The code defines the motion sensor and relay pins, setting one as input and the other as output. The loop continuously checks for motion, and when detected, the Arduino Nano signals the relay to turn on. Fig. 3 shows a code snippet.

Construction and testing

Assemble the circuit on a general-purpose PCB and connect the secondary side of the transformer across the input connector (U1). Once the components are connected as per the circuit diagram,

the device should be installed in a suitable location to ensure that the PIR motion sensor can detect human movement at the main entrance.

Connect the 230V AC, 60W bulb across the output connector (U2) as the test device. Any electrical appliance may be used in place of the bulb to test.

After assembling, enclose the system in a cabinet for protection and proper installation. Once installed, the device can be powered and is ready for use. The primary aim of this device is to automate the control of home appliances and save energy. **EFY**

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